

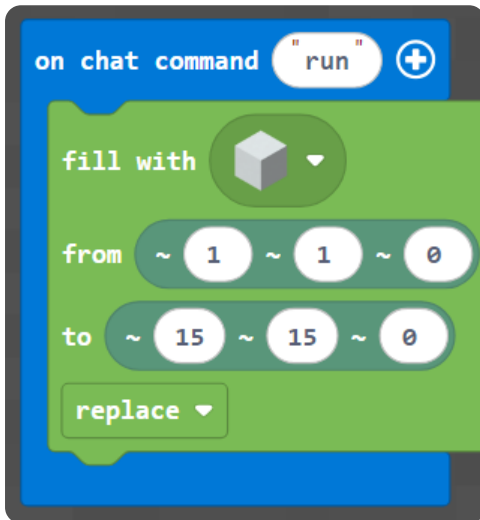
M003 Week 4 — English Worksheet (Beginner)

Topic: Pixel Art 2 — Cat & Tree (x, y) · **Course:** 3D Coordinates · **Level:** Beginner · **Time:** about 30 minutes

This week you copy a **cat** and a **tree**. Each block sits at **(x, y)** — x is **across**, y is **up**.

Build your canvas. Run this to make a blank **15 × 15** wall, then put a **red block at your feet** as your **home spot**:

Blocks



Python

```
def on_run():
    blocks.fill(WHITE_CONCRETE,
                pos(1, 1, 0),
                pos(15, 15, 0),
                FillOperation.REPLACE)
    player.on_chat("run", on_run)
```

● **Stand on the red block every time you run.** If you move, the picture moves too.

1 · Predict 🧠

Read the steps. Circle your answer.

```
place brown block at (8, 1)
place brown block at (8, 2)
place brown block at (8, 3)
```

Across or up? Circle one: across · up

Which number changes? Circle one: x · y

```
place orange block at (2, 15)
place orange block at (3, 15)
```

Across or up? Circle one: across · up

2 · Spot the Bug 🐛

This should make a **tree trunk** going **up** at $x = 8$: (8, 1), (8, 2), (8, 3). But all three blocks land in the **same spot**.

```
place brown block at (8, 1)
place brown block at (8, 1)
place brown block at (8, 1)
```

Hint: to go **up**, the **y** number must grow: 1, 2, 3.

Fill in the blanks to fix it:

```
place brown block at (8, 1)
place brown block at (8, __)
place brown block at (8, __)
```

Which numbers fill the blanks? Circle one: 2 and 3 · 1 and 1

Why was the first code wrong? Circle one: the y never changed · the color was wrong

3 · Show Your Work 📷 🎥

Stand on your **home spot** (the red block).

Warm up first. Copy this small tree. Read each square's (x, y) and place that color there.

 Small tree

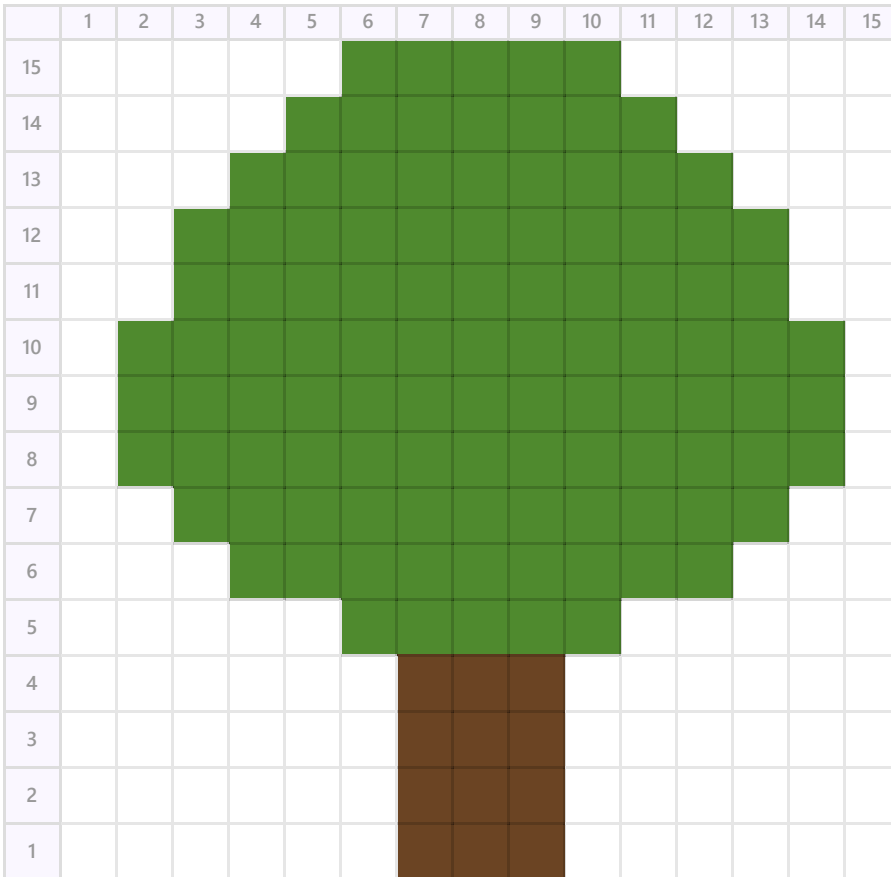
	1	2	3	4	5
5			■		
4		■	■	■	
3			■		
2			■		
1			■		

Try a big one. Pick the 🐱 cat or the 🌳 tree below and copy it the same way. Read each colored square and place that color at its (x, y).

 Cat face

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
15		Orange	Orange										Orange	Orange	
14		Orange	Orange	Orange								Orange	Orange	Orange	
13		Orange	Orange	Orange	Orange						Orange	Orange	Orange	Orange	
12		Orange	Orange	Orange	Orange	Orange	Orange	Orange	Orange	Orange	Orange	Orange	Orange	Orange	
11		Orange	Orange	Orange	Orange	Orange	Orange	Orange	Orange	Orange	Orange	Orange	Orange	Orange	
10		Orange	Orange	Orange	Orange	Orange	Orange	Orange	Orange	Orange	Orange	Orange	Orange	Orange	
9		Orange	Orange		Green	Green				Green	Green				
8		Orange	Orange		Orange	Orange				Orange	Orange				
7		Orange	Orange		Orange	Orange		Pink							
6		Orange	Orange		Orange	Orange		Black							
5		Orange	Orange		Orange	Orange		Orange							
4			Orange	Orange	Orange	Orange	Orange	Orange	Orange	Orange	Orange	Orange	Orange		
3			Orange	Orange	Orange	Orange	Orange	Orange	Orange	Orange	Orange	Orange	Orange		
2				Orange	Orange	Orange	Orange	Orange	Orange	Orange	Orange	Orange			
1						Orange	Orange	Orange	Orange	Orange					

Tree



Record **one video** (a phone is fine). Show two things:

1 • Your code. Scroll through it. Say what each part does.

2 • Your build. Point the camera. Name the parts.

Say these out loud, filling in the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

Submit 

Send this worksheet + your video to teacher on KakaoTalk.